

Question number	Answer	Additional guidance	Mark
6(a)	A description that includes the following points: • increase death rate (with increase in cholesterol) in all countries (except Japan) (1) • increase is greater in United States (and Northern Europe) (1) • death rate higher in Northern Europe (and United States) (1)	ACCEPT positive correlation ACCEPT converse for Mediterranean (and Japan) ACCEPT death rate lowest in Japan (and Mediterranean)	(2)

Question number	Answer	Additional guidance	Mark
6(b)(i)	Two from: headache pins and needles hair loss acne allergic reaction / rash / hives dizziness / low blood pressure memory loss / confusion feeling sick / nausea / vomiting feeling unusually tired / fatigue / physically weak stomach pain digestive system problems, such as constipation, diarrhoea, indigestion or farting loss of appetite (type II) diabetes impotence muscle {pain / tiredness / weakness / damage} joint {pain / swelling} tendon problems sleep problems low blood platelet count / risk of (excessive) bleeding liver damage / hepatitis / yellowing of {eyes / skin} / dark-coloured urine / problems kidney failure / damage / problems		(1)

Question number	Answer	Additional guidance	Mark
6(b)(ii)	An explanation that includes two of the following points: - more {LDL / LDL-cholesterol} will attach to {liver cells / (liver cell) membrane} (1) - therefore more will be {enclosed inside the (membrane) vesicle / (taken) inside the (liver) cell} (in endocytosis) (1)		(2)

Question number	Answer	Additional guidance	Mark
6(c)(i)	- total number of people calculated as 67 500 000 (1) 6.8×10^7 / 6.75×10^7 / 6.750×10^7 (1)	ECF if magnitude wrong in the product using the correct values Bald answer of 6.8×10^7 / 6.75×10^7 = 2 marks Bald answer of 67 500 000 / correct value but incorrect standard form = 1 mark	(2)

Question number	Answer	Additional guidance	Mark
6(c)(ii)	An answer that includes two of the following points: - because statins work by increasing the synthesis of (LDL-)receptors (1) - a homozygous individual will have mutations in <u>both</u> their alleles coding for (LDL-) receptors (1) - therefore no {mRNA / genetic code / instructions} for making the (correct) {receptor / (receptor) protein} (1) - LDL will not be able to bind {to liver cells if no receptor / receptors if wrong shape} (1)	ACCEPT genes (on each chromosome) homozygous individual will have both alleles that are {the same / recessive / dominant} (in LDL-receptor genes) ACCEPT fewer receptors for no receptors	(2)

Question number	Answer	Additional guidance	Mark
6(c)(iii)	An answer that includes the following points: - the {antisense strand of DNA / the drug} will bind to the (apo B) mRNA (1) - preventing translation (so no apo B protein synthesised) (1) OR - transcription of this {DNA / drug} (then translation) will result in a {shortened / different} apo B (protein) (1) - which will {not function properly / less effective / dilute the normal apo B (proteins)} (1)	ACCEPT description of translation ACCEPT description of transcription (shorter) mRNA produced so shorter protein	(2)